



NCCER Module 22105



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Utility Tractors





Section Two

*Starting, Operating, and Shutting Down a
Utility Tractor*

Objective

2. Describe the procedures for starting, operating, and shutting down a utility tractor.
 - a. Describe the safety rules for operating a tractor.
 - b. Identify the daily inspection and maintenance requirements for utility tractors.
 - c. Explain how to properly start up, operate, and shut down a utility tractor.
 - d. Explain how to connect attachments to a utility tractor.



Performance Tasks

Under the supervision of an instructor, trainees should be able to do the following:

1. Perform prestart inspection and maintenance procedures.
2. Properly start, warm up, and shut down a gas-powered and diesel-powered engine tractor.
3. Perform basic maneuvering with a tractor.
4. Attach implements to a drawbar and three-point hitch.
5. Attach and detach implements to a power takeoff.



2.0.0 Starting, Operating, and Shutting Down a Utility Tractor

- Tractors have great potential for causing injury or death to the operator or bystanders if operated incorrectly.
- Operators must have a thorough knowledge of the equipment they are operating, as well as accepted work procedures and safety principles.



2.1.0 Safety Guidelines

Safety should be the number one priority of the operator.

Working safely involves:

- Knowing the proper operation of the machine he or she is using
- Knowing job-site rules and regulations
- Being aware of nearby workers



2.1.1 Safety Rules for Operating Tractors (1 of 3)

- Read the operator's manual thoroughly and know the machine.
- Study the company's safety manual.
- Do not wear loose fitting clothing.
- Wear ear protection if needed.
- Provide for ventilation.
- Become familiar with controls. Be sure to know how to stop the tractor and engine.
- Do not let another person on or near the tractor when it is in operation. With rotating attachments, shut down the tractor if a person or animal approaches.
- Remove or tie down loose tools and equipment.



2.1.1 Safety Rules for Operating Tractors (2 of 3)

- Keep seat belt fastened.
- Stay seated during operation.
- Use all lights and safety warning devices.
- Check brakes.
- Disengage PTO when not operating an attachment.
- Do not smoke while working on the tractor, handling fuel or solvents, or checking or filling batteries.
- Never use attachments that are not matched to tractor size, weight, power, PTO speed, and drawbar or three-point hitch capacity.
- Make sure all safety devices, guards, shields, and safety decals or signs are properly installed as specified by the manufacturer's manual.



2.1.1 Safety Rules for Operating Tractors (3 of 3)

- Make sure all safety devices, guards, shields, and safety decals or signs are properly installed as specified by the manufacturer's manual.
- Observe and follow all safety signs or decals and use any steps or handholds to mount tractor.
- Make sure front and rear frame counterweights, wheel weights, and wheel ballasts are used as recommended by the manufacturer for the attachment being used.
- Never add extra counterweights to compensate for an overload.



2.1.2 Safety Rules for Effective Tractor Operation (1 of 4)

- Perform daily prestart inspection and maintenance procedures.
- Adjust the seat and steering wheel and fasten seat belt before starting the tractor.
- Follow the manufacturer recommended starting procedure.
- After first starting a tractor, quickly check that the controls function properly before moving the tractor.
- Make sure that adequate clearance exists in all directions.
- Make sure that the steering works both left and right and the brakes function properly.
- Make sure that any attachments used do not exceed the load rating of the tractor.



2.1.2 Safety Rules for Effective Tractor Operation (2 of 4)

- Keep all body parts inside the operator's compartment when operating the tractor.
- To prevent the tractor from tipping over backwards, do not lock both rear wheels with the brakes at the same time while the clutch is engaged and the tractor is in a low range and gear with the engine at working speed.
- If both rear wheels become mired (stuck) or frozen and do not move and rotate, do not attempt to move forward. Always use reverse gear to free mired or frozen wheels.
- Keep speed down.
- Avoid sudden starts, stops, and turns. Always slow down before a turn.



2.1.2 Safety Rules for Effective Tractor Operation (3 of 4)

- Use caution when entering the highway.
- Signal all turns and stops.
- Travel completely on the shoulder whenever possible.
- Stay at least 4 to 10 feet (1.2–3 meters) from the edges of deep ditches, embankments, and excavations.
- Do not touch electrical wires with any part of the tractor.
- When turning with a towed attachment connected to the rear of the tractor, make sure that no parts of the tractor or attachment ride up on the curb or each other.
- Park in a non-operating or designated area and on level ground.



2.1.2 Safety Rules for Effective Tractor Operation (4 of 4)

- Before dismounting a tractor, disengage the PTO, lower any attachments to the ground, and retract any hydraulic cylinders, especially single-acting cylinders, as much as possible.
- Shut off the engine and cycle all the SCV control levers through all positions to relieve hydraulic pressure, then return the levers to the neutral position.
- Make sure the fuel tank is full if the tractor will remain unused overnight.
- Follow the company's preventive maintenance program.
- When transporting a tractor, make sure to properly secure it, with the brakes locked and the rain cap covered.



2.1.3 Safety Rules for Operating a Tractor on Slopes or Hills (1 of 2)

- Always reduce speed on slopes or hills.
- If forced to move forward out of a ditch, gully, or up a steep slope, use the lowest gear possible and engage the clutch very slowly. Be prepared to disengage the clutch immediately if the front wheels rise off the ground.
- Avoid driving forward out of deep ditches or gullies. Always back out if possible.
- Go up steep slopes in reverse.
- If work on a slope is unavoidable, go up and down the slope, perpendicular to the toe of the slope.
- Do not go downhill with heavy loads.



2.1.3 Safety Rules for Operating a Tractor on Slopes or Hills (2 of 2)

- Keep the tractor in gear when going downhill.
- Keep all raised equipment close to the ground when working on slopes.
- Make sure the heavy end of the tractor faces uphill.
- Do not cross slopes of more than 25 degrees.
- When crossing steep slopes, avoid turning uphill. If it is unavoidable, slow down and make a wide turn.
- Avoid any depressions with the downhill tires and any bumps, large rocks, stumps, ledges, or raised areas with the uphill tires.



2.1.4 Safety Rules for Operating the Hydraulic System (1 of 2)

- Wear gloves and eye protection when connecting, disconnecting, or checking any high-pressure hoses.
- Before disconnecting high-pressure hydraulic hoses, always relieve pressure as much as possible by first lowering attachments and loads. Retract cylinders as much as possible.
- Shut off the tractor engine and cycle the SCV control levers to relieve pressure.
- Before connecting hoses, make sure the tractor engine is stopped and the appropriate SCV control levers are in the float detent position if connecting hydraulic motors, or in the neutral position if connecting hydraulic cylinders.
- Never use the SCV control lever to adjust hydraulic motor speeds.



2.1.4 Safety Rules for Operating the Hydraulic System (2 of 2)

- Never place SCV control levers in float detent before lowering the load.
- Always check that the hydraulic fluid level does not fall below the minimum level with all single-acting cylinders in the fully extended position.
- Stand clear of equipment when working with hydraulic lines and tools.
- Keep the hydraulic system clean.
- Use the proper fittings for hoses and attachments.
- Never depend on the hydraulic system to support machine parts.
- Avoid prolonged contact with hydraulic fluid.



2.1.5 Safety Rules for Operating the Power Takeoff (1 of 2)

- Make sure the attachment is compatible with the PTO.
- Keep PTO guards and shields in place.
- Always disconnect the PTO when not in use.
- Never engage the PTO while the machine engine is shut off.
- Keep hands, feet, and clothing away from the PTO parts.
- Never step or cross over a PTO part.
- Do not wear loose clothing.
- Never operate the PTO shafts at extreme angles.
- Never ride, or let others ride, on the drawbar of the tractor.



2.1.5 Safety Rules for Operating the Power Takeoff (2 of 2)

- Be sure the PTO spinner shields rotate freely at all times.
- Disengage all power and shut off the tractor engine before checking spinner shields.
- Always be sure the PTO stub shaft is properly secured to the machine's power shaft.
- Do not service, lubricate, or perform other work without first disengaging the PTO and shutting off the tractor engine.

PTO WARNING DECAL
ON PTO SHIELD



2.1.6 Safety Rules for Tractor Tires

- Check tires daily for damage or noticeably low pressure.
- Check inflation pressure with an accurate gauge.
- Inflation pressures of less than 12 psi (83 kPa) should be monitored.
- Tire pressure can be increased by 4 psi (28 kPa) but traction will be reduced.
- Tire pressures of tractors operating on side slopes or hills should be increased by 4 psi (28 kPa).
- Never stand parallel to the rim when inflating a tire.
- Stand clear of the tire and rim if lifting with a chain or cable sling.
- Never weld or perform torch work on wheels with inflated tires.



2.2.0 Daily Inspection and Maintenance

- Preventive maintenance of the equipment is important.
- Each day before beginning operation of the tractor, perform a walk-around inspection and any prescribed daily maintenance checks.



2.2.1 Servicing a Tractor

The following may be needed to service a tractor:

- air supply
- cleaning solvents
- engine coolant
- engine oil
- fuel
- grease gun
- hydraulic fluid
- lubricants
- operator's manual
- rags
- tire pressure gauge
- tool kit
- transmission fluid
- water



2.2.2 Typical Daily Inspection and Maintenance Procedures (1 of 2)

- Follow the company's preventive maintenance program when servicing equipment.
- Visually inspect for leaks and broken, worn, or missing parts.
- Report major repairs and replacements to the proper authority.

Service Interval Chart-Daily or 10 Hours/50 Hours/First 100 Hours/First 300 Hours/300 Hours					
ITEM	Daily or 10 Hours	50 Hours	First 100 Hours	First 300 Hours	300 Hours
Check Engine Oil Level	*				
Drain Water and Sediment from Fuel Tank and Fuel Filter	*				
Check Engine Coolant Level		*			
Check Transmission-Hydraulic System Oil Level		*			
Check MFWD Axle Housing and Wheel Hub Oil Level		*			
Inspect Tires and Check Inflation Pressure		*			
Lubricate Adjustable Front (2WD) Axle Steering Spindles and Cylinder Ends		*			
Lubricate MFWD Steering Kingpins, Axle Pivot Pin and Rear Axle	* *	*			
Inspect Tractor for Loose Hardware		*			
Change Engine Oil and Filter			*		
Replace Transmission-Hydraulic Filter			*		
Check Clutch Pedal Free Play			*		
Change Engine Oil and Filter (5225 and 5325)				*	
Service Engine Air Cleaner		* *			*
Change Engine Oil and Filter					*
Lubricate Hitch Components					*
Check Neutral Start System					*
Check Clutch Pedal Free Play					*
Clean Cab Air Filters		* *			*
Adjust PTO Clutch Lever Linkage					*

*Service if operated in extremely dusty or wet conditions.



2.2.2 Typical Daily Inspection and Maintenance Procedures (2 of 2)

- Check all fluid levels. Add fluids if necessary.
- Check the filters, hose, lines, and fittings and look for leaks. Replace if necessary.
- Clean and lubricate grease fittings according to operator's manual.
- Check the pressure and condition of the tires.
- If operating in extremely dusty or trashy conditions, check the grille, radiator, and air conditioning condenser, if so equipped, for dirt or trash buildup.



2.3.0 Startup, Shutdown, and Operation of a Diesel or Gasoline Engine

Before attempting any prestart checks or startup, always read the operator's manual for the tractor to become familiar with specific controls and procedures.



2.3.1 Prestart Checks

- Before starting or moving a tractor, perform any required daily inspection, maintenance, and safety checks.
- Make sure that the tractor's fuel tank is full.
- Check that all other persons are clear of the tractor.
- Observe pinch points and obstructions.



2.3.2 Startup (1 of 2)

- Mount the tractor while maintaining three points of contact.
- Adjust the seat and fasten the seat belt if the tractor has a ROPS.
- Make sure that the tractor is set in Park or that the parking brake is set.
- Check that the PTO, hitch, and any Hi/Lo shift and reverser controls are in the Off or Neutral position.
- Make sure that SCV controls are in Neutral; however, if a hydraulic motor is connected to any SCV, make sure that it is set to Float.
- Once the controls are set, check all warning/option indicator lights.
- Set the throttle to the one-third or one-half position and, for a diesel, turn on the fuel supply if required.



2.3.2 Startup (2 of 2)

- Press the clutch if required and activate the starter to crank the engine.
- Release the starting device when the engine starts.
- If a choke is set, push it in slowly as the engine warms up.
- Check that warning lights and gauges register the correct readings.
- Release the clutch while the engine warms up.
- Observe the temperature gauge to determine when the engine has reached the proper operating temperature.
- If the tractor has power steering, turn the wheels back and forth occasionally to determine if the hydraulic fluid is warm enough to allow easy steering.
- In very cold weather, consult the operator's manual for any procedures to allow rapid warming of the hydraulic fluid.



2.3.3 Shutdown

- Shift the transmission to Park or set the parking brake and shift the transmission to Neutral.
- Lower attachments to the ground and retract hydraulic cylinders.
- Set the PTO, hitch, and any Hi/Lo shift and reverser controls to the Off or Neutral position.
- Reduce the engine speed. For diesel engines, turn off the fuel, if required. Next, turn off the ignition.
- Cycle all SCV controls through all positions and then back to neutral.
- Turn off all switches.
- If the tractor will be parked overnight, make sure that the fuel tank is full.



2.3.4 Safely Stopping a Tractor

WARNING:

Prepare to stop far in advance and look for possible hazards.
Avoid sudden stops.

- Depress the clutch (if present) or reduce forward speed and gradually apply both brakes.
- If the tractor will be stopped for a period of time with the engine running, shift the transmission to Park or set the parking brake and shift the transmission to Neutral.
- Set the throttle position to about one third for gasoline engines or one half for diesel engines for cooling purposes.

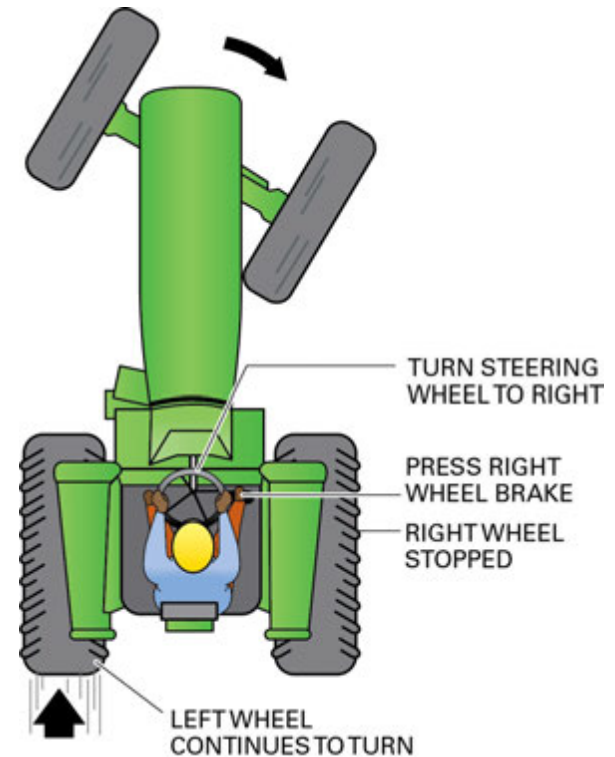


2.3.5 Basic Tractor Operations

Shifting procedures vary depending on transmission type:

- Standard (non-synchronized)
- Synchronized
- Power-assisted
- Hydrostatic

Left or right turns are made by turning the steering wheel while moving and applying a corresponding rear wheel brake.



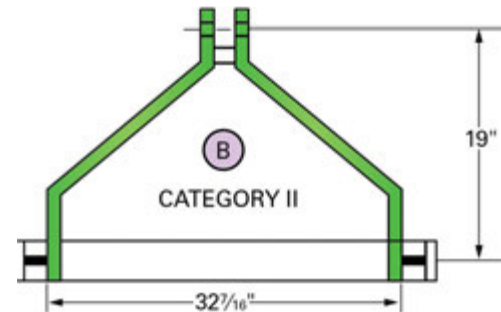
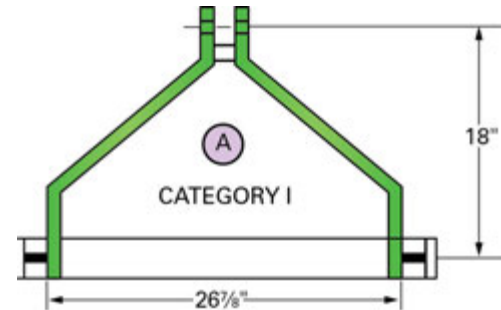
2.4.0 Connecting Attachments

- A three-point hitch is the standard hydraulic linkage for tractor attachments.
- Three-point hitches consist of two lower links and an adjustable top link.



2.4.1 Preparing the Attachment

- A three-point attachment hitch is narrower and is used for smaller attachments than Category II (B) attachments.
- When connecting a Category I attachment to a Category II tractor hitch, install reducer bushings in the links to accommodate the smaller pins.



A – CATEGORY I
B – CATEGORY II

2.4.2 Connecting the Attachment (1 of 4)

1. Check the operator's manual for attachment installation instructions.
2. Set the rockshaft or hitch control lever to the minimum position.
3. Check the drawbar position to make certain it will not interfere with movement of the hitch.
4. Back the tractor so that draft links are in position to connect to attachment hitch pins.



2.4.2 Connecting the Attachment (2 of 4)

5. Raise the draft links with the rockshaft unit or hitch control lever to the height needed to connect to attachment hitch pins.
6. Place the transmission selector lever in Park or Neutral.
7. Connect the left draft link to the left hitch pin of the attachment. Always attach the left draft link first.
8. Insert the locking pin.
9. Raise or lower the right draft link to the correct height by adjusting the lift link.
10. Attach the right draft link and insert the pin.



2.4.2 Connecting the Attachment (3 of 4)

11. Unhook the center link from the transport bracket.
12. Adjust the center link to reach the mast by turning the outer housing or by using the adjusting handle.
13. Adjust the center link so that both shafts inside the outer housing are extended equally.



2.4.2 Connecting the Attachment (4 of 4)

14. Drop the center link into the attachment mast and lock into place by inserting and fastening a pin through holes in the mast and turnbuckle. If the center link is equipped with a latch assembly, depress the lever, fit the latch over the ball, and release the lever.
15. Adjust sway bars or chains or install sway blocks in a lower position with the narrow, flat edge inward. Be sure that the draft links do not bind with the sway blocks.
16. Install any front-end balance weights, as specified for the attachment.
17. Raise the attachment to the transport position, using the rockshaft or hitch control lever. Make certain that it clears at all points and that the connections are secure.



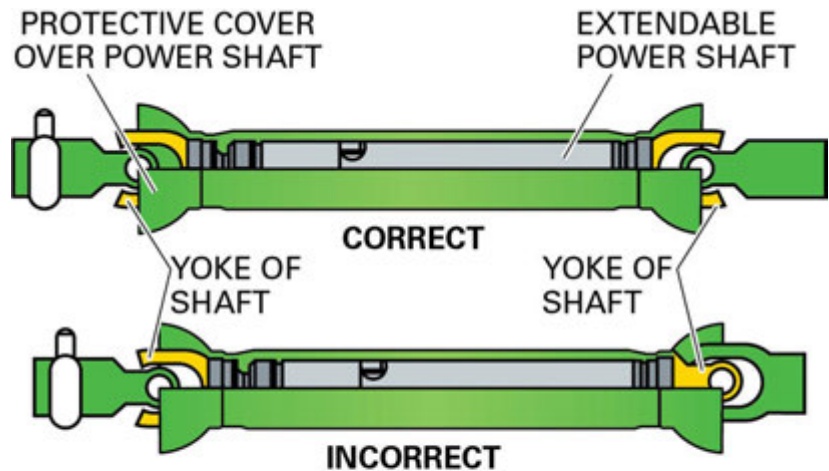
2.4.3 Attaching a PTO (1 of 4)

1. Be sure to have the correct rpm PTO stub shaft for the attachment being connected.
2. Set the wheel brakes.
3. Disengage the power to the PTO shaft.
4. Shut off the engine.
5. Remove the PTO shaft guard or master shield if the tractor is so equipped.
6. Check the tractor PTO coupling and the attachment PTO stub shaft to make sure they have the same number of splines.



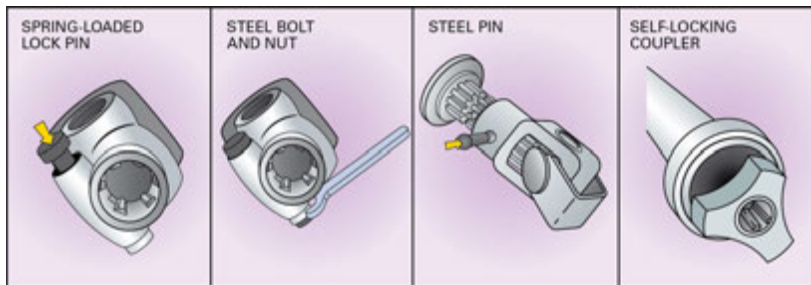
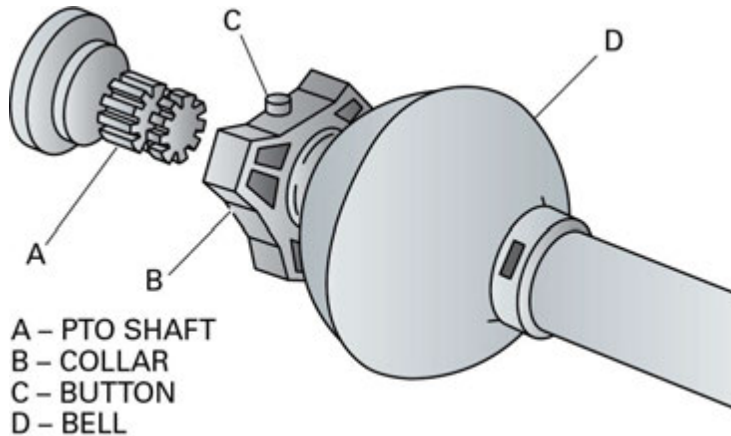
2.4.3 Attaching a PTO (2 of 4)

7. Clean the inside of the coupling of all grease and debris.
8. Oil the PTO shaft spline.
9. Verify that the yokes on the shaft are aligned identically as shown.



NOTE POSITION OF RIGHT HAND SHAFT YOKES
Make sure that the yokes of the shaft are aligned along the same plane.

2.4.3 Attaching a PTO (3 of 4)



10. Attach the power shaft with a spring-loaded lock, if applicable. Other types of locking devices are common, as shown below.

11. Move the collar back against the stop on the yoke.

12. Turn the collar to align the yoke with splines on the tractor PTO shaft.

13. Slide the power shaft onto the PTO shaft.

2.4.3 Attaching a PTO (4 of 4)

14. Depress the pin or button and move the shaft forward until the collar snaps into the locked position.
15. Check the locking device by pulling back on the bell or guard shaft. Do not pull on the collar; the latch will release.
16. Slide the safety shield into place.
17. Keep the alignment as straight as possible.
18. Maintain the proper drawbar height and length.
19. Reinstall the PTO master shield.



2.4.4 Operating a PTO

- The PTO should always be disengaged when not operating the attachment.
- Start the tractor engine and, if required, depress the clutch pedal and both brakes all the way.
- Engage the PTO and, if applicable, release the brakes and engage the clutch slowly.
- To shift gears and leave the PTO running, depress the clutch and, if required, both brakes. Change gears and, if used, release the brakes. Engage the clutch slowly.
- To disengage the PTO and, if required, stop the tractor to do so, depress the clutch and both brakes. When the tractor stops, disengage the PTO.



2.4.5 Connecting and Disconnecting Hydraulic Hoses for Attachments (1 of 4)

WARNING: Always wear gloves and eye protection when connecting, disconnecting, or checking any high-pressure hoses.

To connect hoses:

1. Stop the tractor engine.
2. Cycle the SCV control levers through all positions. Return levers to the float position if connecting hydraulic motors or to the neutral position if connecting hydraulic cylinders.



2.4.5 Connecting and Disconnecting Hydraulic Hoses for Attachments (2 of 4)

3. Remove and save any attachment hose nipple caps. Then wipe the nipples clean.
4. Open the covers on the SCV couplers and verify that they are clean.
5. Select the proper coupler for a single-acting or double-acting cylinder extend and the corresponding hose nipple. Insert the nipple in the coupler and push in firmly to **seat** the nipple.
6. Repeat Step 5 for the retract coupler and hose for double-acting cylinders.
7. Pull all hoses slightly to make sure they are seated in the couplers.



2.4.5 Connecting and Disconnecting Hydraulic Hoses for Attachments (3 of 4)

8. Start the tractor engine and move the SCV control levers from neutral to extend and retract the cylinders slightly; or from float to full retract and back to float to start and stop hydraulic motors. Then, check for signs of any visible leaks.
9. For hydraulic motors, adjust the motor speed as described previously.



2.4.5 Connecting and Disconnecting Hydraulic Hoses for Attachments (4 of 4)

To disconnect hoses:

1. Lower all loads/attachments to the ground and retract all cylinders as much as possible.
2. Stop the tractor engine, cycle the SCV control levers through all positions to relieve pressure at the couplings, and return the levers to the Neutral position.
3. Pull the hoses from the SCV couplers.
4. Replace the attachment hose nipple caps and close the SCV coupler caps.



Wrap Up – Trade Terms

Bead: When speaking of inflatable tires, the rigid inner ring of a tire that fits against rim and creates an air-tight seal when the tire is inflated.

Seat: To cause two surfaces to fit firmly together.



2.0.0 Section Review Question 1

1. To keep a tractor from tipping backwards, _____.
 - a. lock both rear wheels with the brakes at the same time when moving forward in low range
 - b. do not attempt to move forward if the rear wheels are mired
 - c. attach loads as high as possible with the hitch
 - d. drive out of ditches by moving forward



2.0.0 Section Review Question 1 Answer

1. To keep a tractor from tipping backwards, _____.
 - a. lock both rear wheels with the brakes at the same time when moving forward in low range
 - b. do not attempt to move forward if the rear wheels are mired**
 - c. attach loads as high as possible with the hitch
 - d. drive out of ditches by moving forward



2.0.0 Section Review Question 2

2. Which of the following is NOT a typical interval for the associated service item?
- a. Change engine oil and filter every 300 hours.
 - b. Clean cab air filters every 50 hours in dusty conditions.
 - c. Check tire inflation pressure every 50 hours.
 - d. Drain water and sediment from fuel tank and filter every 300 hours.



2.0.0 Section Review Question 2 Answer

2. Which of the following is NOT a typical interval for the associated service item?
- a. Change engine oil and filter every 300 hours.
 - b. Clean cab air filters every 50 hours in dusty conditions.
 - c. Check tire inflation pressure every 50 hours.
 - d. Drain water and sediment from fuel tank and filter every 300 hours.**



2.0.0 Section Review Question 3

3. When first starting a tractor, make sure that _____.
a. the PTO is engaged
b. all SCVs are energized
c. the transmission is in Park
d. the transmission is in Neutral with the parking brake off



2.0.0 Section Review Question 3 Answer

3. When first starting a tractor, make sure that _____.
a. the PTO is engaged
b. all SCVs are energized
c. the transmission is in Park
d. the transmission is in Neutral with the parking brake off



2.0.0 Section Review Question 4

4. When connecting an attachment, the draft links are raised with the _____.
- a. rockshaft unit
 - b. throttle lever
 - c. drawbar
 - d. SCV lever



2.0.0 Section Review Question 4 Answer

4. When connecting an attachment, the draft links are raised with the _____.
- a. rockshaft unit**
 - b. throttle lever
 - c. drawbar
 - d. SCV lever



Next...



Module Review and Examination

Complete the Review Questions at the end of the module and prepare for the Module Exam.